MAPLE & MAIN · CUSTOMER SEGMENTATION

Five personas worth acting on, and the four customers in five we cannot label yet

A needs-based segmentation that is reproducible, stable and checked against what customers actually bought next, with an honest account of where it does not reach.

Synthetic data. No client, employer or proprietary information.

EXECUTIVE SUMMARY

The personas are real. The problem is how few customers have enough history to earn one.

1.66x

more of next year's spending lands in a persona's own aisles than the average customer's. Personas fixed in May 2025, purchases measured over the following year.

19%

of active customers have a basket clear enough to label. Two years of history instead of one raises that to 33%.

2.2 points

difference in offer response between the five personas, against a margin of error of ±3.8. Personas tell you what to say, not who to contact.

Recommendation: use personas for merchandising and message, not for audience selection; label on two years of purchases rather than one; and leave the 81% of customers without a clear persona on general content rather than guessing.

APPROACH

Two questions, kept apart, and four ways to break the answer

"What does this customer want?" and "how do they like to shop?" are different questions. One clustering of every variable answers only the second.

1. Build

Personas from the aisles a customer buys from. Channel habit is kept as a separate attribute.

SQL · 96,371 active customers

2. Assign with care

A persona is given only when the basket explains it at least 7 times better than the average shopper.

Multinomial likelihood

3. Test

Reproducible on a fresh sample? Stable a year later? Or just value tiers with new names?

Three tests

4. Prove it is useful

Does the label predict next year's shopping, and does it change what we should do?

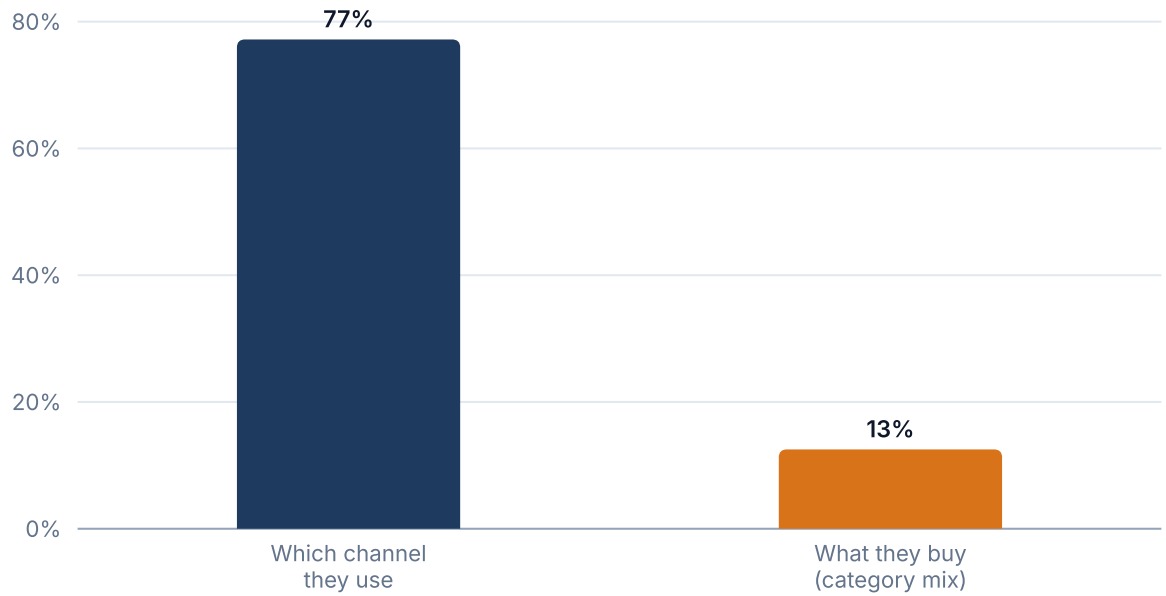
Out-of-time and campaign holdouts

DATA USED	CHECKS BEFORE MODELLING	HOW RESULTS WERE VALIDATED
Twelve months of orders and order lines for every active customer, measured twice a year apart	Category and channel shares verified to add up; unit counts reconciled to totals; no missing model inputs	Split-half test, twelve-month stability, next-year purchases, and the latent shopper types the data was generated from

WHY ONE CLUSTERING IS NOT ENOUGH

Put every variable in one model and it hands back a channel report

What a single clustering of all behaviour separates



Share of the variation in each kind of behaviour that the groups explain. Five groups, all 22 behaviour measures.

GROUP	WHAT DEFINES IT
33% of customers	Orders in store high, Orders on web low
24% of customers	Orders on web high, Orders in store low
15% of customers	Average basket low, Categories bought in a year low
14% of customers	Electronics share high, Average basket high
14% of customers	Orders in app high, Orders in store low

Channel use is close to all-or-nothing, so it dominates the distances. The result is useful for contact strategy and useless for merchandising, which is the question that was asked.

THE PERSONAS

Five needs-based groups, named after the aisles they over-index on

Sports
13% of labelled customers

Spend a year	$2,357
Orders	10.9
Items	37
Gold tier	34%
Typical age	35-44

Electronics
24% of labelled customers

Spend a year	$2,796
Orders	10.8
Items	36
Gold tier	34%
Typical age	35-44

Grocery & Toys
17% of labelled customers

Spend a year	$1,798
Orders	12.5
Items	42
Gold tier	41%
Typical age	35-44

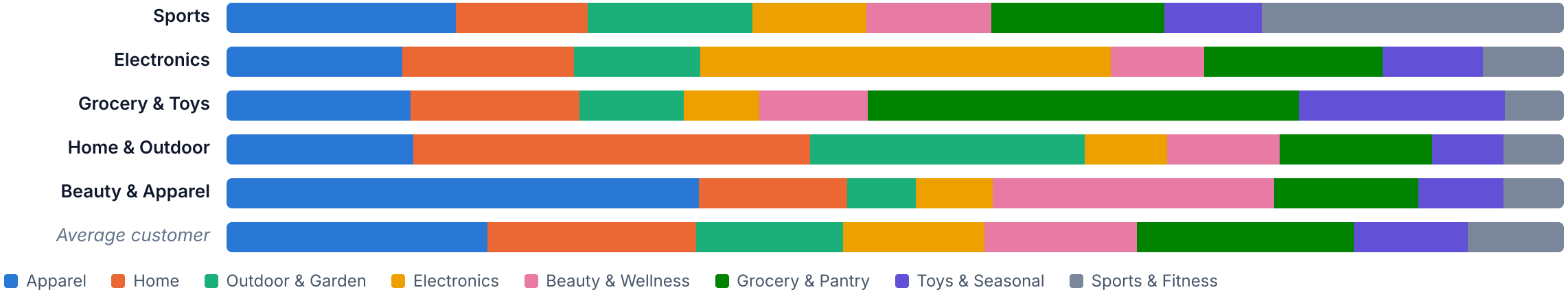
Home & Outdoor
20% of labelled customers

Spend a year	$2,462
Orders	12.3
Items	42
Gold tier	42%
Typical age	35-44

Beauty & Apparel
26% of labelled customers

Spend a year	$2,027
Orders	11.9
Items	40
Gold tier	38%
Typical age	25-34

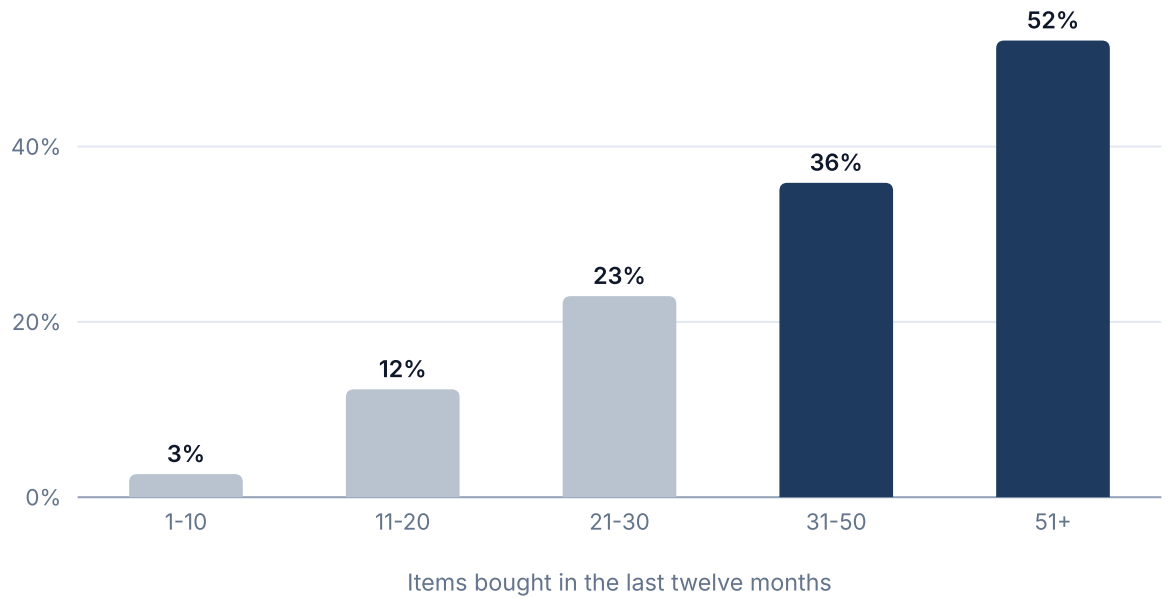
Where each persona's units come from, against the average customer



EVIDENCE

A label is only as good as the basket behind it

Share of customers given a persona, by items bought in the year



A customer is labelled only when one persona explains their basket at least 7 times better than the average shopper.

The median active customer buys 18 items a year across six categories. That is not enough to tell a Home shopper from someone who bought a kettle.

Rather than label everyone and hope, the rule leaves 81% of customers unlabelled. They are not a persona called "Other": they are customers we do not yet know.

> **The cheapest fix is more history, not a better algorithm.** Looking back two years instead of one lifts the labelled share from 19% to 33%, because the median customer then has 34 items on file.

THE THREE TESTS

Reproducible, stable, and not value tiers in disguise

1. Reproducible

0.95

Agreement between segmentations built independently on two halves of the customer base. The same five groups appear both times.

Adjusted Rand index, where 1 is identical and 0 is chance.

2. Stable

82%

of customers labelled in both years keep the same persona, against 21% if labels were shuffled. Classic RFM segments keep only 31%.

Same customers, segmented in May 2025 and May 2026.

3. Not value tiers

5%

of the variation in spend is explained by persona, against 52% for RFM. Every persona contains high and low spenders.

Which is the point: value belongs in a score, not a persona.

> RFM is the opposite trade: it explains half the variation in spend and almost none of the variation in what people buy (1%, against 38% here). The two answer different questions and both have a place.

CHECKED AGAINST THE TRUTH

Where the labels are confident, three in four are right

Latent shopper type of the customers in each persona

PERSONA	MATCHING SHOPPER TYPE	SHARE
Beauty & Apparel	Fashion & Beauty	79%
Home & Outdoor	Home & Garden	79%
Grocery & Toys	Family Everyday	77%
Electronics	Tech & Gadgets	75%
Sports	Active Outdoors	61%

The synthetic data was built from five hidden shopper types. A real project never sees these, which is what makes this check worth doing here.

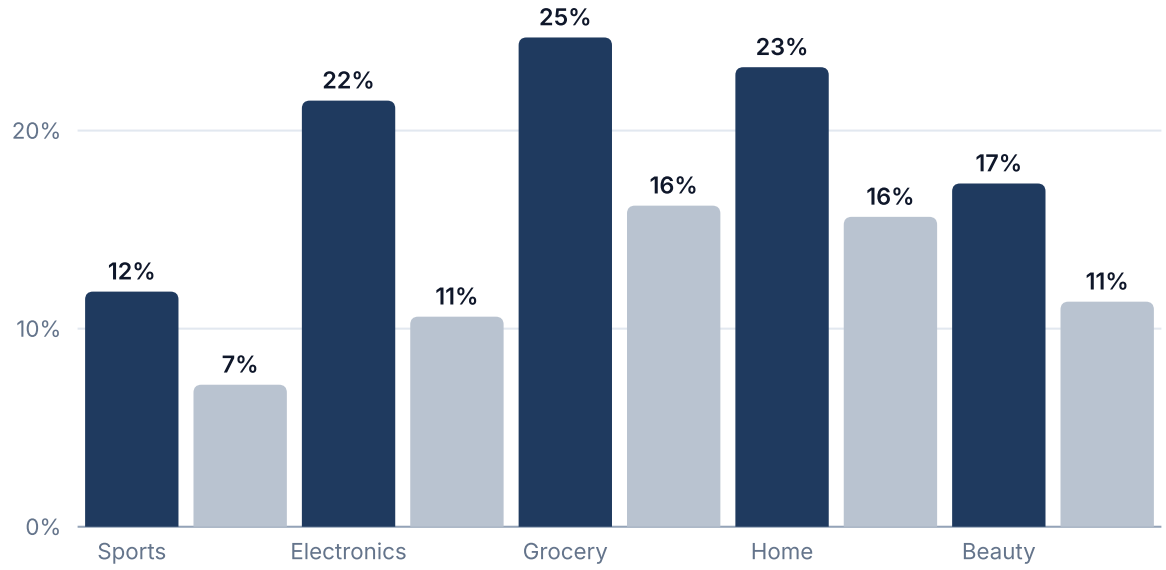
- **75%** of confidently labelled customers belong to the type their persona points at. Chance would give 20%.
- Agreement with the hidden types is **0.50** among labelled customers and **0.01** if every customer is force-labelled.
- Customers who get a label have genuinely stronger tastes: average type strength 0.54 against 0.46.

> The stability test and the truth agree. That matters: on real data only the stability test is available, and this says it can be trusted.

DOES IT PREDICT ANYTHING?

A persona points at the right aisle a year later. An unlabelled customer does not.

Share of next year's items bought in the persona's own aisle



Dark bars: customers with that persona. Light bars: the average customer. Personas fixed in May 2025; purchases measured over the following twelve months for 69,790 customers.

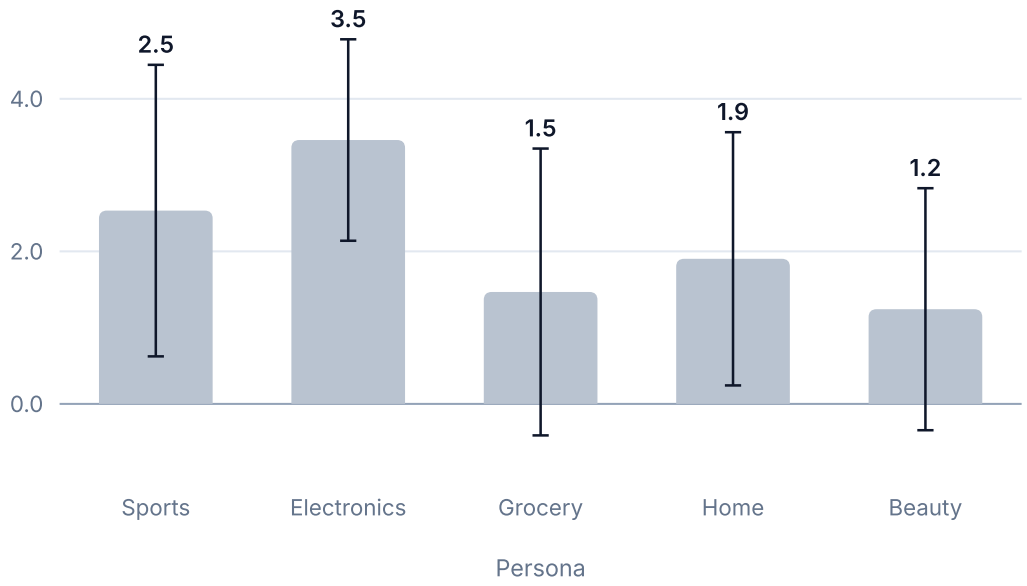
PERSONA	NEXT YEAR'S TOP AISLE IS THE PERSONA'S	BASE RATE
Sports	9%	3%
Electronics	37%	9%
Grocery & Toys	46%	17%
Home & Outdoor	44%	19%
Beauty & Apparel	14%	7%

For customers left unlabelled, next year's spending lands in each aisle at **100%** of the base rate: exactly no signal. The confidence rule is separating knowledge from noise, not throwing away data.

WHAT PERSONAS DO NOT DO

They do not tell you who to contact, and they are scarcest where you need them

Extra customers who order because of an offer, by persona



Percentage points, treated minus holdout, 13 campaigns over twelve months. Bars show 95% intervals.

The gap between the best and worst persona is **2.2 points**, with a margin of error of ±3.8. On this evidence, persona is not a basis for choosing an audience. Project 1 answers that question.

	HAS A PERSONA	AVERAGE CHURN RISK
All active customers	21%	14%
Top 30% by value at risk	**17%**	**n/a**
Customers with a persona	100%	8%

> Labels come from buying a lot, so the customers we know best are the ones least likely to leave. Among the customers project 1 would contact first, only 17% can be personalised this way.

RECOMMENDATIONS

Four actions

1 **Use personas for content, not for audience**

Pick who to contact with the value-at-risk score from project 1, then let the persona choose the aisle in the message. A persona-led offer is 2.3x more relevant than a generic one.

No new data needed

2 **Label on two years of purchases**

The single cheapest improvement: it nearly doubles the share of customers who can be personalised, with no change to the method.

19% → 33% of customers

3 **Leave unlabelled customers on general content**

Guessing a persona for them would be guessing: their next-year buying matches the average exactly. Browsing and wish-list data would help here more than any model.

Protects 81% from bad targeting

4 **Test the persona-led message against a holdout**

Relevance is measured; incremental revenue is not. Run persona-led content against a random-content control for a quarter before rolling it out.

Decides the roll-out

APPENDIX

Method, assumptions and limits

Population. Customers of at least a year's standing with two or more orders in the twelve months before the snapshot: 96,371 at May 2026.

Personas. K-means on category unit shares, shrunk toward the average shopper by 20 items so a small basket cannot look like a specialist. Five groups, chosen for reproducibility and stability rather than fit.

Assignment. Each customer's items are scored against every persona and against the average shopper, using a multinomial likelihood. The persona is given only when it beats the runner-up by 7 times.

Channel. Treated as a separate attribute: store-first, web-first, app-first, or no single channel. In this data it is uncorrelated with persona, so the two carry different information.

Fit statistics are weak, and that is expected. The silhouette score is 0.14. Shopping behaviour has no natural gaps; segments are built, not discovered. Reproducibility, stability and prediction are the tests that matter.

Limits. Relevance is measured; the revenue effect of acting on it is not. Personas describe the past year and lapse when a customer buys less: of those labelled in 2025, 35% still qualified in 2026.

Synthetic data. Maple & Main is fictional. Its customers were generated from five latent shopper types with varying strength, which is why the personas can be scored for accuracy. Figures illustrate the method, not any real company.

Code. SQL, Python and SAS, with a segmentation card and QC log, are in the project repository.